Skills Summary

Signed Rational Numbers: Choosing the Right Move

Use this reference while completing your worksheet or when studying.

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1. Adding and subtracting signed numbers

Rewrite every subtraction as an addition: $p - q = p + (-q)$, so $-6 - (-11)$ becomes $-6 + 11$.

Then look at the signs. Same signs: add the distances from zero and keep the sign. Different signs: subtract the smaller distance from the larger and keep the sign of the number further from zero.

Example: Evaluate $-5 - (-9)$.
Step 1. Subtraction with a negative behind it is the shape that causes sign slips, so rewrite it as an addition before deciding anything.
Step 2. $-5 + 9$: the signs differ, 9 is further from zero, so the answer is positive with size $9 - 5$.
$-5 - (-9) = \mathbf{4}$

Try it: Evaluate $-7 - (-2)$.

check: -5

2. Multiplying and dividing signed numbers

Work out the sizes first, then fix the sign by **counting the negative factors**: an even count gives a positive answer, an odd count a negative one. The same rule covers division, because dividing is multiplying by the reciprocal.

Example: Evaluate $(-6)(-7)$.
Step 1. Both numbers are negative, so the size and the sign can be settled separately — multiply the sizes, then count the minus signs.
Step 2. $6 \times 7 = 42$, and there are two negative factors, an even count, so the product is positive.
$(-6)(-7) = \mathbf{42}$

Try it: Evaluate $-45 \div 9$.

check: -5

3. Comparing rationals in different forms

Put both numbers in the **same form** first — dividing the fraction out to a decimal is usually quickest. Then use the number line: further left is smaller, so among negatives, the one *further from zero* is the smaller number.

Example: Write $<$ or $>$ between $-\frac{2}{5}$ and -0.5.
Step 1. A fraction and a decimal cannot be compared by eye, so convert one of them and compare like with like.
Step 2. $-\dfrac{2}{5} = -0.4$, and 0.4 is closer to zero than 0.5, so -0.4 sits to the right of -0.5.

$$-\frac{2}{5} > -0.5$$

Try it: Write $<$ or $>$ between $-\frac{7}{8}$ and -0.85.

check: $<$

4. Mixed expressions: the order that decides

Brackets, then exponents, then multiplication and division **left to right**, then addition and subtraction left to right. Equal-rank operations are never swapped: $-24 \div (-6) \cdot (-2)$ divides first.

Example: Evaluate $-2 - 3(-4)$.
Step 1. Two operations of different rank are competing, so the multiplication has to be finished before the subtraction can happen.
Step 2. $3(-4) = -12$, so the expression is $-2 - (-12) = -2 + 12$.

$$-2 - 3(-4) = \mathbf{10}$$

Try it: Evaluate $-5 + \frac{6^2}{-9}$.

check: $\mathbf{-9}$